cl_px4_mr::CbPx4PathFollowerBase
+ CbPx4PathFollowerBase()
+ ~CbPx4PathFollowerBase()
+ onStateOrthogonalAllocation()
+ setFollowerParams()
+ followerParams()
+ onEntry()
+ onExit()
+ update()
# buildPath()
# behaviorName()
# onPathStarted()
# onPathCompleted()
# totalLength()
# progressFraction()
# carrotVertexIndex()
- commandFor()
- applyCentering()
- tangentYawAt()
cl_px4_mr::CbPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~CbPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()
weak_ptr< smacc2::introspection::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection::SmaccStateInfo >
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
int
+stateMachine_
+parentState_
+transitions_
+children_
+tid_
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+depth_
+stateIndex_
SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2::introspection::SmaccStateInfo >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
recursive_mutex
atomic< StateMachineInternalAction >
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
vector< smacc2_msgs::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
-reactorWalkDepth_
#nh_
#timer_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
+demangledStateName
+fullStateName
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
mutex
-mutex_
-stateMachine_
cl_px4_mr::PathFollowerParams
-lastProgressDecile_
+yawMode
+arrivalXyTol
+arrivalZTol
+autoTimeoutFactor
+fixedYaw
+groundSpeed
+leash
+minSegmentLength
-onEntryThread_
-onExitThread_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
-isShutdownRequested_
smacc2::SmaccAsyncClientBehavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
+x
+y
+yaw
+z
cl_px4_mr::NedPoint
-onFailure_
-onFinished_
-onSuccess_
-heading_
-refAlt_
-x_
-y_
-z_
+onArmed_
+onDisarmed_
+onLanded_
+onModeChanged_
-lastUpdate_
+onLocalPositionReset
+onPositionReceived_
-armingState_
-navState_
-prevArmingState
-prevNavState_
-xyResetCounter_
-zResetCounter_
-periodDuration_
-projection_
-goalActive_
-hasPublished_
cl_px4_mr::CpVehicleStatus
+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()
-goalX_
-goalY_
-goalZ_
-xyTolerance
-zTolerance_
+onGoalReached_
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
cl_px4_mr::CpVehicleLocalPosition
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
cl_px4_mr::CpVehicleCommand
+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
TrajectorySetpoint
cl_px4_mr::TunnelCenteringParams
+ minRangeM
+ wallBandHalfHeightM
+ verticalSearchM
+ wallFadeM
+ smoothing
+ceilingClearanceM
+columnHalfWidthM
+floorClearanceM
+floorLookBackM
+lateralSearchM
+lookAheadMaxM
+lookAheadMinM
+maxLateralM
+maxVerticalM
+minFloorClearanceM
-enabled_
-localPosition_
-lastSetpoint_
cl_px4_mr::CpGoalChecker
+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
cl_px4_mr::CpTrajectorySetpoint
+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()
-trajectorySetpoint_
-params_
-completed_
-active_
-entryHeading_
-sCarrot_
-totalLen_
#followerParams_
-lastCmd_
#vehicleStatus_
#localPosition_
#vehicleCommand_
#goalChecker_
#trajectorySetpoint_
cl_px4_mr::CpOffboardKeepAlive
+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()
optional< std::chrono::steady_clock::time_point >
-firstMessage_
-initialized_
-end_
-initialized_
-rosInitialized_
+queueSize
+postInitialMessageEvent
+postMessageEvent
-sub_
-lastState_
+onFirstMessageReceived
+onMessageReceived_
-updatableClients_
-updatableStateElements_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
smacc2::client_core_components::CpTopicSubscriber< MessageType >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
function< void(const sensor_msgs::msg::PointCloud2 &)>
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
-haveAttitude_
+queueSize
+onFirstMessageReceived
+onMessageReceived_
-lastCloudNs_
< sensor_msgs::msg::PointCloud2 >
+postInitialMessageEvent
+postMessageEvent
smacc2::client_core_components::CpTopicSubscriber< sensor_msgs::msg::PointCloud2 >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
array< float, 4 >
atomic< float >
time_point
-params_
-timeoutMs_
-subscriber_
-attitudeQ_
-ceiling_
-floor_
-lateral_
-left_
-right_
-vertical_
-lastLog_
cl_px4_mr::CpTunnelCentering
+ CpTunnelCentering()
+ ~CpTunnelCentering()
+ onInitialize()
+ lateralOffsetM()
+ verticalOffsetM()
+ valid()
+ leftWallM()
+ rightWallM()
+ floorM()
+ ceilingM()
+ params()
- onCloud()
- onAttitude()
-lastUpdateTime_
#tunnelCentering_
vector< cl_px4_mr::NedPoint >
vector< float >
atomic< size_t >
-path_
-cumLen_
-carrotVertex_